KEY SHEET

CHEMISTRY

1)	**A**	2)	**A**	3)	**D**	4)	**B**	5)	**A**	6)	**B**	7)	**D**	8)	**BD**
9)	**ABD**	10)	**ABC**	11)	**AD**	12)	**C**	13)	**A**	14)	**B**	15)	**B**	16)	**D**
17)	**5**	18)	**8**	19)	**5**	20)	**0**	21)	**4**	22)	**2**	23)	**6**		

PHYSICS

24)	**A**	25)	**B**	26)	**B**	27)	**A**	28)	**B**	29)	**B**	30)	**C**	31)	**AB**
32)	**BC**	33)	**ABD**	34)	**AC**	35)	**B**	36)	**B**	37)	**C**	38)	**D**	39)	**C**
40)	**0**	41)	**5**	42)	**7**	43)	**8**	44)	**5**	45)	**2**	46)	**9**		

MATHEMATICS

47)	**A**	48)	**B**	49)	**D**	50)	**A**	51)	**A**	52)	**D**	53)	**D**	54)	**ABD**
55)	**CD**	56)	**ABD**	57)	**ABC**	58)	**A**	59)	**C**	60)	**D**	61)	**C**	62)	**D**
63)	**9**	64)	**7**	65)	**8**	66)	**1**	67)	**5**	68)	**0**	69)	**9**		

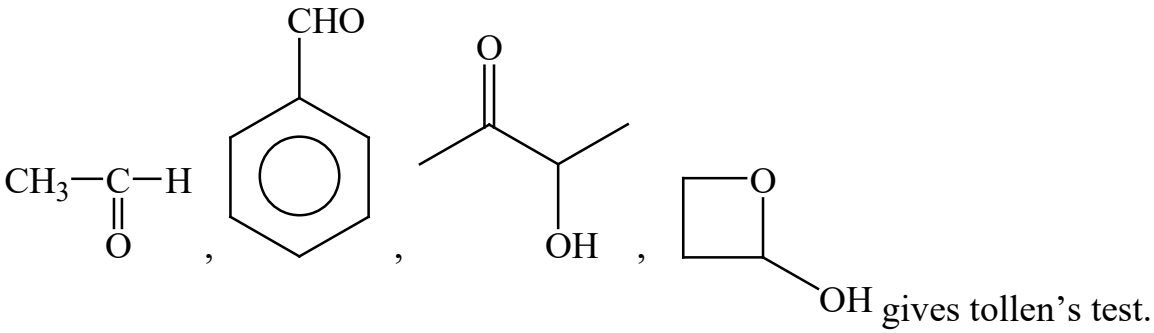
SOLUTIONS
CHEMISTRY

1. $acac^{-1}$: form six membered ring

2. $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

3. I-4, R; II-3, Q; III-2, S; IV-1, P

4.



gives tollen's test.

5. Hydrolysis of PO_4^{-3} will take place

$$\underset{(e-x)}{PO_4^{-3}} + H_2O \rightleftharpoons \underset{(x-y)}{HPO_4^{-2}} + \underset{(x+y+z)}{OH^-} \left[k_{h_1} = \frac{k_w}{k_3}\right]$$

$$\underset{(x-y)}{HPO_4^{-2}} + H_2O \rightleftharpoons \underset{(y-z)}{H_2PO_4^-} + \underset{(x+y+z)}{OH^-} \left[k_{h_2} = \frac{k_w}{k_2}\right]$$

$$\underset{(y-z)}{HPO_4^-} + H_2O \rightleftharpoons \underset{z}{H_3PO_4} + \underset{(x+y+z)}{OH^-} \left[k_{h_3} = \frac{k_w}{k_1}\right]$$

$$\left[OH^-\right] = x + y + z \qquad \left[OH^-\right] = x$$

$x >> y >> z$

$$k_{h_1} = \frac{(x+y+z)(x-y)}{(c-x)} \qquad \frac{k_w}{k_3} = \frac{x.x.}{(c-x)}$$

$$2.2 \times 10^{-2} = \frac{x^2}{(0.1-x)}$$

On solving $x = 3.73 \times 10^{-2}$ molar

6. CONCEPTUAL

7. CONCEPTUAL

8. (i) $XeF_4 + H_2O \rightarrow Xe + HF + \underset{(para)}{O_2} + XeO_3$

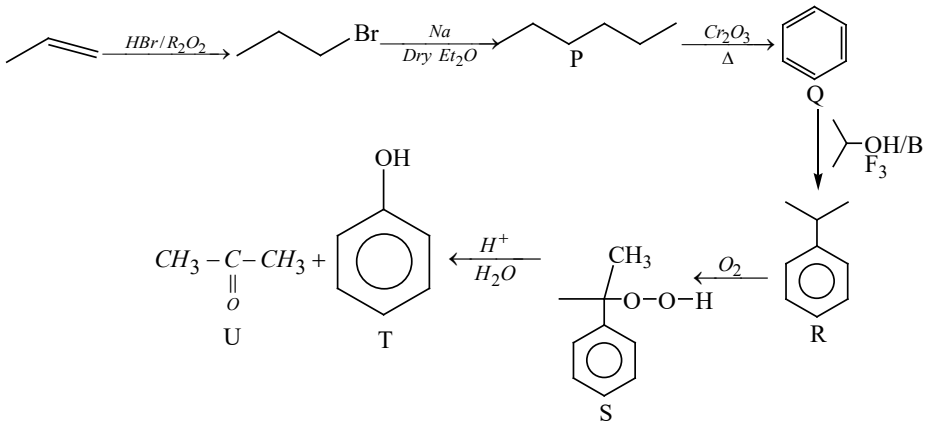
(ii) $Zn + HNO_3 \rightarrow N_2O + Zn(NO_3)_2 + H_2O$

(iii) $H_2O_2 + I_2 \xrightarrow{OH^-} I^- + \underset{(para)}{O_2}$

(iv) $XeF_6 + H_2O \rightarrow XeO_3 + HF$

9. $S - S > O - O$

10.



11.

$$3A \longrightarrow 2B + 5C$$

10 0 0 at $t = 0$

(10-5) $\frac{3}{5}\times\frac{10}{3}$ $\frac{2}{5}\times\frac{25}{3}$ at $t = t_{1/2}$

Mole of $C = \frac{2}{5}\times\frac{25}{3} = \frac{10}{3}$ Mole of $B = \frac{3}{5}\times\frac{10}{3} = 2$

12 & 13:

S = Benzoil alcohol

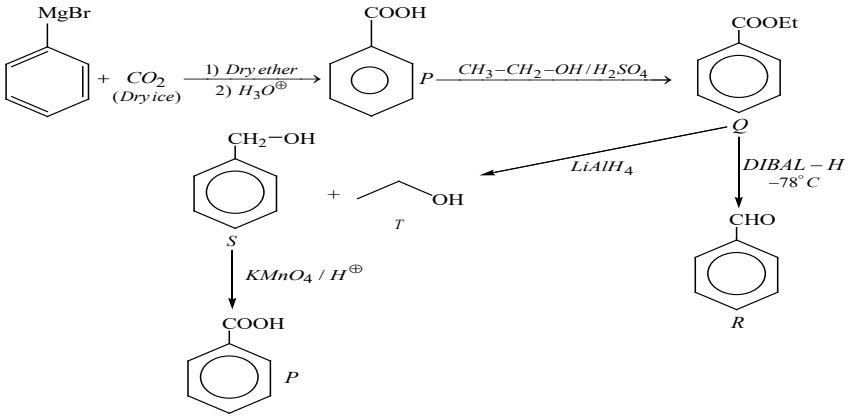
T = Ethanol

P =Benzoic acid

Q = Ethylbenzoate

R = Benzaldehyde

14.



15. Mole of $NH_4HS = \frac{306}{51} = 6$

Mole of H_2S at equilibrium $= \frac{170}{34} = 5$

$$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$$

6 O O

1 5 5 – at equilibrium

At equilibrium- $P = \frac{10\times 0.0821\times 500}{4.1}$

P = 100 atm $\quad P_{NH_3} = 50\ atm \quad P_{H_2S} = 50\ atm$

For the new equilibrium

$$NH_4HS(s) \rightleftharpoons \underset{x}{NH_3(g)} + \underset{100\ atm}{H_2S(g)}$$

$K_P = x \times 100 \quad 2500 = x \times 100 \quad x = 25$

16. $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g) \quad .....(i)$

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g) \quad .....(ii)$

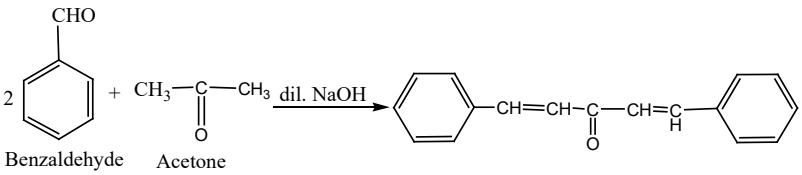
For reaction (i) K_p remains unchanged because temp in constant but total pressure will increase.

17.

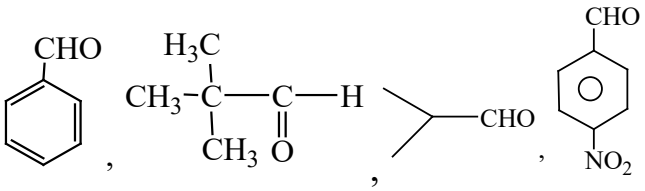
18. $Si_3O_{10}^{8-}$

19. $B_2H_6,\ BeH_2(solid),\ (AlH_3)_n,\ \left[Be(CH_3)_2\right]_n,\ Al_2(CH_3)_6$

20.



21.



22.

	Mole of A	Mole of B
In liquid	x	(20 – x)
In vapours	(10 – x)	x

$$\frac{y_A}{y_B} = \frac{x_A}{x_B}\frac{P_A^\circ}{P_B^\circ} \qquad \frac{10-x}{x} = \frac{x}{(20-x)}\frac{P_A^\circ}{P_B^\circ}$$

$(10-x)(20-x) = x^2\dfrac{P_A^\circ}{P_B^\circ}$ On solving we get

$X = 7.25 \qquad X_A = \dfrac{7.25}{20} = 0.36 \quad X_B = 0.64$

$P = X_A P_A^\circ + X_B P_B^\circ \qquad = 0.79$ atm

23. $\dfrac{d}{P.f} = \dfrac{3M}{4\pi R^3.N_A} \quad d = 6\ gram/cm^3$

PHYSICS

24. $T = 2\pi\sqrt{\dfrac{I}{mgx}}$

25. $v_p = -v\,x$ slope

$x = \pi/8$ at $t = \pi/6$

Particle is at mean position

26. % error $= 100\left(\dfrac{\Delta l}{l} + \dfrac{4\Delta r}{r}\right)$

$= \left(\dfrac{0.1}{10} + 4\left(\dfrac{0.01}{2}\right)\right)\times 100 = 3$

27. According to photo electric effect equation:

$KE_{\max} = \dfrac{hc}{\lambda} - \phi_0$

$\dfrac{p^2}{2m} = \dfrac{hc}{\lambda} - \phi_0$

$\dfrac{(h/\lambda_d)^2}{2m} = \dfrac{hc}{\lambda} - \phi_0$

Assuming small changes, differentiating both sides,

$\dfrac{h^2}{2m}\left(-\dfrac{2d\lambda_d}{\lambda_d^3}\right) = -\dfrac{hc}{\lambda^2}d\lambda$

$\dfrac{d\lambda_d}{d\lambda} \propto \dfrac{\lambda_d^3}{\lambda^2}$

28. $J_c = \sigma E_0 \sin(\omega t - kx)$

$i_{d=\in\in_0\frac{d\phi_E}{dt}=\in\in_0 A\frac{dE}{dt}}$

$=\in\in_0 \times A\ E_0\omega\cos(\omega t - kx)$

$Jd =\in\in_0 E_0\omega$

$\dfrac{J_c}{J_d} = \dfrac{\sigma}{\in\in_0\ \omega}$

29. In rod, $\dfrac{\lambda}{4} = l \Rightarrow \lambda = 4l \Rightarrow v_r = \lambda v = 4lv$

In air column $v_a = 2\Delta l v; \dfrac{v_r}{v_a} = \dfrac{2l}{\Delta l} \Rightarrow v_r = \dfrac{2\times 1\times 330}{0.6} = 1100\,m/s$

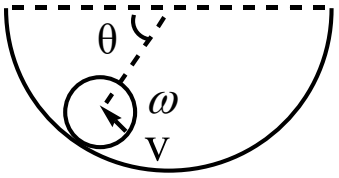
30. Conceptual

31. Conceptual

32. $mgL(\sin\alpha - \sin\beta) = \dfrac{1}{2}MV^2 + \dfrac{1}{2}I\omega^2$

33. flux through inductor remain constant

34. Magnetic forces due to angular motion of sphere will not create any torque. Magnetic forces due to translational motion passes through centre of mass of sphere acts parallel to normal reaction



$v = r\omega$ .....(1)

$$\frac{1}{2}\left(\frac{2}{5}+1\right)mr^2\left(\frac{v}{r}\right)^2 = mg(R-r)\sin\theta$$

$$v = \sqrt{\frac{10}{7}g(R-r)\sin\theta} \quad .....(2)$$

$$m\frac{v^2}{(R-r)} = N - mg\sin\theta - qvB$$

In limiting condition $N = 0; \theta = 90^0$

$$qvB = mg + \frac{10}{7}mg = \frac{17mg}{7}$$

$$B_{\text{limiting}} = \frac{17mg}{7qv} = 2T$$

35. Net torque about centre of mass of system is zero, therefore angular momentum of system will remain conserved .

36. Conserving angular momentum about centre of rod

$$\frac{ML^2}{12}\omega_0 = \left[\frac{ML^2}{12} + 2m\left(\frac{1}{2}t\right)^2\right]\omega \Rightarrow \omega = \frac{L^2\omega_0}{(L^2+t^2)} \Rightarrow \int_0^\theta d\theta = L^2\omega_0\int_0^t \frac{dt}{L^2+t^2}$$

$$\theta = L\omega_0 \tan^{-1}\left(\frac{t}{L}\right)$$

Since path of centre of mass is parabolic

$\therefore$ time after which centre will return to same horizonal level is $T = \frac{2\times 10\sin 30}{g} = 1\text{sec}$

$$\therefore \theta = 8\left(\tan^{-1}(1)\right) = 2\pi$$

37. Insects will reach ends after time 1 sec.

Angular velocity of rod at that instant is $\omega = \frac{L^2\omega_0}{L^2+t^2} = \frac{8}{2} = 4\ rad/s$

Friction force $= f = \frac{1}{6}\times\frac{1}{2}\times 16 = \frac{4}{3}N$

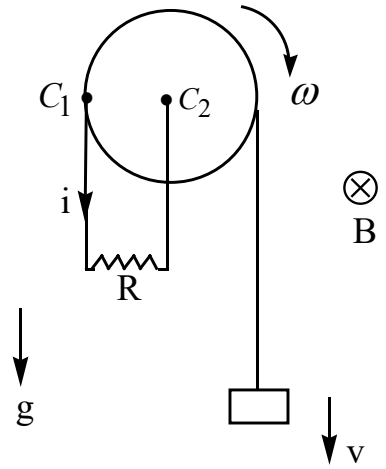
38. Writing energy equation

$$mgv = mv\frac{dv}{dt} + I\frac{\omega d\omega}{dt} + i^2R \quad .....(i)$$

$v = a\omega$

$$\frac{dv}{dt} = a\frac{d\omega}{dt}$$

$$i = \frac{B\omega a^2}{2R} = \frac{Bva}{2R} \qquad \ldots\ldots(ii)$$



From equation (i) and (ii)

$$\frac{3m}{2}\frac{dv}{dt} = mg - \frac{B^2a^2}{6mR}$$

$$v = \frac{4mgR}{B^2a^2}\left(1 - e^{-\left(\frac{B^2a^2}{6mR}\right)t}\right)$$

$$\frac{dv}{dt} = \frac{2g}{3}e^{\left(\frac{B^2a^2}{6mR}\right)}$$

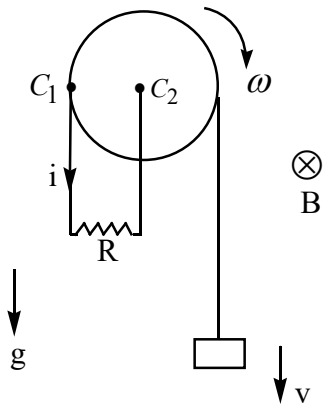
39. Writing energy equation

$$mgv = mv\frac{dv}{dt} + I\frac{\omega d\omega}{dt} + i^2R \qquad \ldots\ldots\ldots(i)$$

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From equation (i) and (ii)

$$\frac{3m}{2}\frac{dv}{dt} = mg - \frac{B^2a^2}{6mR}v$$

$$v = \frac{4mgR}{B^2a^2}\left(1 - e^{-\left(\frac{B^2a^2}{6mR}\right)t}\right)$$

$$\frac{dv}{dt} = \frac{2g}{3}e^{\frac{B^2a^2}{6mR}t}$$

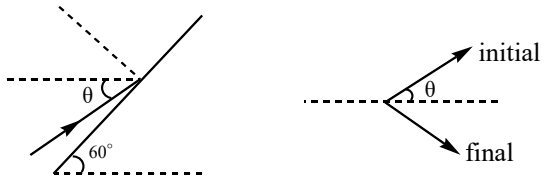
40. Path length travelled by rays on reaching S_1 and S_2 from S is same, so the control maximum will be formed on the central line.

41. $r_1 = r_2 = 30^0$

$$1(\sin i_1) = \sqrt{2}\left(\frac{1}{2}\right) = \frac{1}{\sqrt{2}}$$

$i_1 = 45^0$

$\theta = 15^0$



Charge in momentum of each photon $= 2P\sin(15^0)$

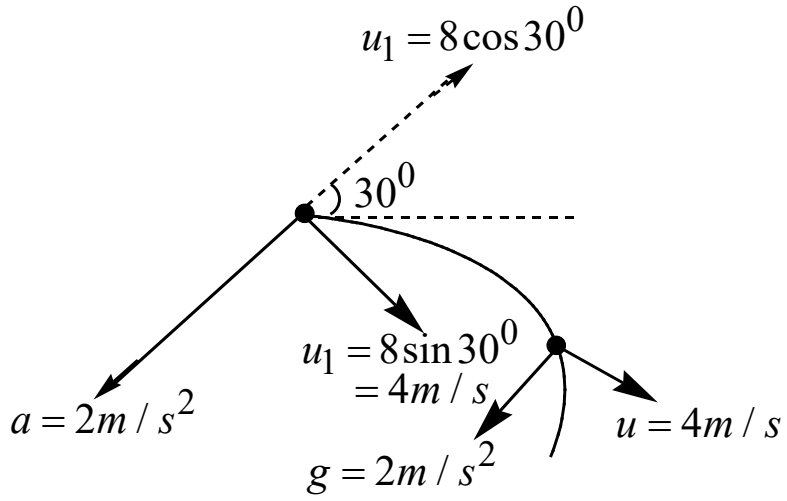
$$\text{Force} = \left(\frac{P_0}{P_c}\right)2P\sin(15^0) = \frac{2P_0}{c}\sin 15 = \frac{P_0}{c}\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right) \quad x = 3 \qquad y = 2$$

42. $mv + M(u) = (M+m)2$

$M + mv = (M+m)u$

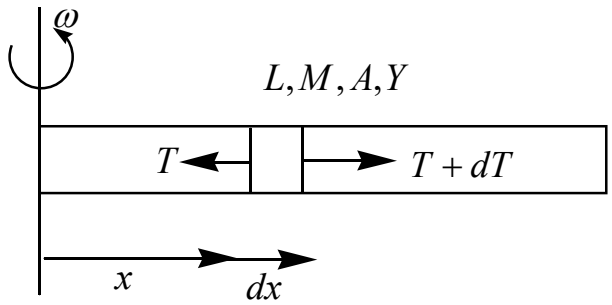
43. The acceleration vector shall change the component of velocity u along the acceleration vector $r = \frac{v^2}{a_0}$. Radius of curvature $r_{\min}$ means v is minimum and a_0 is maximum .This is

at point P when component of velocity parallel to acceleration vector becomes zero, that is u=0



$$\therefore R = \frac{u_1^2}{a} = \frac{4^2}{a} = 8meter$$

44.



$$dl = \frac{M\omega^2}{2AL}\left(L^2 - x^2\right)dx$$

$$\frac{\Lambda l_0}{\Delta l_i} = \frac{\int_{L/2}^{L}\left(L^2 - x^2\right)dx}{\int_{0}^{L/2}\left(L^2 - x^2\right)dx} = \frac{5}{11}$$

45. From the given information $Z = 2$. The energy of photon is 10.2eV. so, threshold energy is 6.2eV

46. use Q value of nuclear reaction

MATHEMATICS

47. $\sin x = 2 + [\sin x] + [-\cos x]$ .....(i)

$\{\sin x\} = 2 + [-\cos x]$

$0 \le \{\sin x\} < 1$

and $2 + [-\cos x]$ can have values 1,2,3

Hence no solution

48. $x^2 \dfrac{dy}{dx} - xy = 1 + \cos\left(\dfrac{y}{x}\right)$ or $\dfrac{x(xdy - ydx)}{dx} = 1 + \cos\left(\dfrac{y}{x}\right)$

$$\Rightarrow \frac{\dfrac{(xdy - ydx)}{x^2}}{1 + \cos\left(\dfrac{y}{x}\right)} = \frac{dx}{x^3} \Rightarrow \int \frac{d\left(\dfrac{y}{x}\right)}{1 + \cos\left(\dfrac{y}{x}\right)} = \int \frac{dx}{x^3} \Rightarrow \frac{1}{2}\int \frac{d\left(\dfrac{y}{x}\right)}{\cos^2\left(\dfrac{y}{2x}\right)} = \int \frac{dx}{x^3}$$

$$\Rightarrow \frac{1}{2}\frac{\tan\left(\dfrac{y}{2x}\right)}{\dfrac{1}{2}} = \frac{x^{-2}}{-2} + c \Rightarrow \tan\left(\frac{y}{2x}\right) + \frac{1}{2x^2} = c$$

49. Put $\dfrac{2x}{3} = \tan\theta$

Then $2\sin^{-1}(\cos 2\theta) + 3\cos^{-1}(\sin 2\theta) + \tan^{-1}\tan 2\theta = \lambda\pi$

$$\Rightarrow 2\left(\frac{\pi}{2} - \cos^{-1}\cos 2\theta\right) + 3\left(\frac{\pi}{2} - \sin^{-1}\sin 2\theta\right) + \tan^{-1}\tan\theta = \lambda\pi$$

For $x = \dfrac{-17}{13} \to \dfrac{-\pi}{2} < 2\theta < 0$ $\quad \therefore\ 2\left(\dfrac{\pi}{2} + 2\theta\right) + 3\left(\dfrac{\pi}{2} - 2\theta\right) + 2\theta = \lambda\pi$

$$\frac{5\pi}{2} = \lambda\pi \Rightarrow \lambda = 2.50$$

50. Let $I = \displaystyle\int_{\pi/4}^{\pi/2} \frac{dx}{\cos x(\tan x + 1 + 2\sqrt{\tan x})\sqrt{\tan x \cos^2 x}}$

$$I = \int_{\pi/4}^{\pi/2} \frac{\sec^2 x dx}{(1 + \sqrt{\tan x})^2 \sqrt{\tan x}}$$

Put $\tan x = t^2 \Rightarrow \sec^2 x dx = 2t\,dt$

$$I = \int_1^{\infty} \frac{2t\ dt}{(t+1)^2 . t} \qquad I = -2\left[\frac{1}{t+1}\right]_1^{\infty} = -2\left[0 - \frac{1}{2}\right] = 1$$

51. {1,2,3,4}, {1,2,4,3}, {1,3,2,4}, {2,1,3,4}, {2,1,4,3}

52. We note that $\left|f(a+ib)\right| = \sqrt{e^{2a}\cos^2 b + e^{2a}\sin^2 b} = e^a > 0$. In particular, $f(a+ib) \neq 0$ and f is not onto. Since cos and sin are 2π – periodic, f is not one – to – one:

$f\left(a + i\left(b + 2\pi\right)\right) = e^a\left(\cos\left(b + 2\pi\right) + i\sin\left(b + 2\pi\right)\right) = e^a\left(\cos\left(b\right) + i\sin\left(b\right)\right) = f\left(a + ib\right).$

53. We take the logarithm and perform the following transformations:

$$-\frac{3}{2}\ln n+\sum_{k=1}^{n}\frac{n+k}{n^2}\ln(n+k)=-\frac{3}{2}\ln n+\frac{1}{n}\sum_{k=1}^{n}\left(1+\frac{k}{n}\right)\ln\left(n\left(1+\frac{k}{n}\right)\right)=$$

$$=-\frac{3}{2}\ln n+\frac{1}{n}\sum_{k=1}^{n}\left(1+\frac{k}{n}\right)\left[\ln n+\ln\left(1+\frac{k}{n}\right)\right]=$$

$$=\ln n\left[-\frac{3}{2}+\frac{1}{n}\sum_{k=1}^{n}\left(1+\frac{k}{n}\right)\right]+\frac{1}{n}\sum_{k=1}^{n}\left(1+\frac{k}{n}\right)\ln\left(1+\frac{k}{n}\right)=$$

$$=\frac{\ln n}{2n}+\frac{1}{n}\sum_{k=1}^{n}\left(1+\frac{k}{n}\right)\ln\left(1+\frac{k}{n}\right)=\int_1^2 x\ln x\,dx\,.$$

Hence, $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{k=1}^{n}\left(1+\frac{k}{n}\right)\ln\left(1+\frac{k}{n}\right)=\int_1^2 x\ln x dx=2\ln 2-\frac{3}{4}$

54. Let $P(A\cup B)=x \;\Rightarrow P(A)=\frac{2x}{3},P(B)=\frac{x}{2}$ &

$$P(A\cap B)=\frac{2x}{6}.P(B)=\frac{x^2}{6} \qquad \Rightarrow x=\frac{2x}{3}+\frac{x}{2}-\frac{x^2}{6}$$

$$\Rightarrow x=P(A\cup B)=1 \;\&\; P(A\cap B)=\frac{1}{6}$$

55. $f(x)-e^x+\frac{1}{f(x)}-e^{-x}=0$

$$(f(x)-e^x)+\frac{e^x-f(x)}{e^x f(x)}=0$$

$$(f(x)-e^x)(e^x f(x)-1)=0 \qquad \Rightarrow f(x)=e^x \text{ or } e^{-x}$$

$$\text{or } \begin{cases} e^x, & x\ge 0 \\ e^{-x}, & x<0 \end{cases} \quad \text{or} \quad \begin{matrix} e^x, & x<0 \\ -e^{-x}, & x\le 0 \end{matrix}$$

$$\int_0^1 e^x dx=e-1$$

$$\int_0^1 e^{-x}dx=1-\frac{1}{e}$$

$\therefore$ Area can be $e-1+e-1$ or $1-\frac{1}{e}+1-\frac{1}{e}$

56. (A) $\frac{(a_3-2d)+(a_3-d)+a_3+(a_3+d)+(a_3+2d)}{5}$

$$>\left((a_3-2d)(a_3-d)(a_3)(a_3+d)(a_3+2d)\right)^{1/5}$$

$$\Rightarrow a_3^5>(a_2^3-4d^2)(a_3^2-d^2)a_3$$

$\Rightarrow a_3^4 > a_3^4 - 5a_3^2d^2 + 4d^4 \Rightarrow 5a_3^2 > 4d^2$

(B) $a_1 + a_2 + a_3 + a_4 + a_5$

$a_1 + a_5 = 2a_3$ & $a_2 + a_4 = 2a_3$

(C) Not possible

(D) $(a_3 - 2d)\ (a_3 + 2d) < (a_3 - d)\ \ (a_3 + d)$

$\Rightarrow d^2 > 0$ which is true.

57. Let P $(at^2, 2at)$

Normal at P passes through centre of circle

$y = -tx + 2t + t^3$ pass through (2,8) $\Rightarrow t = +2$

Point $P(4,4)$

Slope normal = -2

Slope of tangent $= \frac{1}{2}$

58. $P(\overline{w}) = P(E_1 \cap \overline{w}) + P(E_2 \cap \overline{w})$

$= P(E_1).P(\overline{w}) / E_1) + P(E_2).P(\overline{w} / E_2)$

$= \frac{1}{3}.\frac{9}{10} + \frac{2}{3}.\frac{8}{10} = \frac{25}{30} = \frac{5}{6}$

59. $P(\overline{s}) = P(E_1 \cup \overline{s}) + P(E_2 \cap \overline{s})$

$= P(E_1).P(\overline{s} / E_1) + P(E_2).P(\overline{s} / E_2)$

$= \frac{1}{3}.\frac{9}{10} + \frac{2}{3}.\frac{9}{10} = \frac{26}{30} = \frac{13}{15}$

60. $P\left(\frac{E_1}{\overline{w}}\right) = \frac{P(E_1).P(\overline{w} / E_1)}{P(E_1).P(\overline{w} / E_1) + P(E_2).P(\overline{w} / E_2)} = \frac{\frac{1}{3}.\frac{9}{10}}{\frac{1}{3}.\frac{9}{10} + \frac{2}{3}.\frac{8}{10}} = \frac{9}{25}$

61 & 62 paragraph:

$z^{12} + z^6 + 1 = 0$ ……..(i)

$(z^6 - 1)(z^{12} + z^6 + 1) = 0$

$z^{18} - 1 = D$

Let roots of the above equations are $z_1, z_2, z_3, ...z_{18}$

$z = (1)^{1/18}$

$z = (\cos 2n\pi + i \sin 2n\pi)^{1/18}$

$= \cos\frac{2n\pi}{18} + i\sin\frac{2n\pi}{18}$

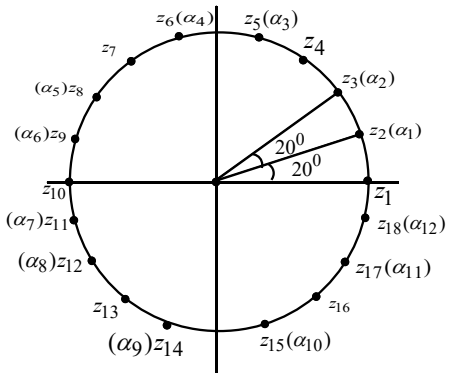
Where, $n - 0,1,2,.....,17$

$z_1 = \cos 0^0 + i \sin 0^0 = 1$

$z_2 = \cos\frac{2\pi}{18} + i\sin\frac{2\pi}{18} = \cos 20^0 + i \sin 20^0$

$$z_3 = \cos\frac{4\pi}{18} + i\sin\frac{4\pi}{18} = \cos 40^0 + i\sin 40^0$$

$$z_{18} = \cos\frac{34\pi}{18} + i\sin\frac{34\pi}{18} = \cos 340^0 + i\sin 340^0$$



$$z_6 - 1 = 0$$

$$z = \cos\frac{2n\pi}{6} + i\sin\frac{2n\pi}{6},\ n = 0,1,2,3,4,5$$

$$z'_1 = \cos 0^0 + i\sin 0^0 = z_1$$

$$z'_2 = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3} = \cos 60^0 + i\sin 60^0 = z_4$$

$$z'_3 = \cos\frac{4\pi}{6} + i\sin\frac{4\pi}{6} = \cos 120^0 + i\sin 120^0 = z_7$$

$$z'_4 = \cos\frac{6\pi}{6} + i\sin\frac{6\pi}{6} = \cos 180^0 + i\sin 180^0 = z_{10}$$

$$z'_5 = \cos\frac{8\pi}{6} + i\sin\frac{8\pi}{6} = \cos 240^0 + i\sin 240^0 = z_{13}$$

$$z'_6 = \cos\frac{10\pi}{6} + i\sin\frac{10\pi}{6} = \cos 300^0 + i\sin 300^0 = z_{16}$$

$\therefore$ Roots of the equation $z^{12} + z^6 + 1 = 0$

are $z_2, z_3, z_5, z_6, z_8, z_9, z_{11}, z_{12}, z_{14}, z_{15}, z_{17}, z_{18}$ which are respectively $\alpha_1, \alpha_2, \alpha_3, ...\alpha_{12}$

61. $\cos\left(\sum_{i=1}^{12}\theta_i\right) + i\sin\left(\sum_{i=1}^{12}\theta_i\right) = \cos(\theta_1 + \theta_2 + ... + \theta_{12}) + i\sin(\theta_1 + \theta_2 + ... + \theta_{12})$

$= e^{i\theta_1}e^{i\theta_2}e^{i\theta_3}....e^{i\theta_{12}} = \alpha_1\alpha_2\alpha_3...\alpha_{12} = 1$

(product of the roots of the equation (i))

62. $\alpha_7, \alpha_8, \alpha_9, \alpha_{10}, \alpha_{11}, \alpha_{12},$ are the roots whose imanginary parts are negative.

Sum of their real parts $= \cos 200^0 + \cos 220^0 + \cos 240^0 + \cos 260^0$

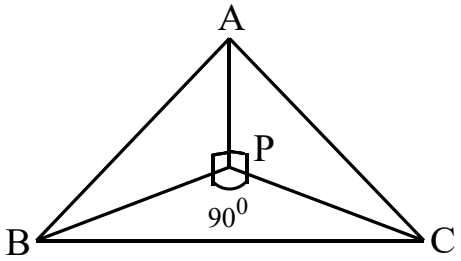
$+ \cos 280^0 + \cos 300^0 + \cos 320^0 + \cos 340^0$

$- \cos 240^0 - \cos 300^0$

$= \frac{\sin 8.10}{\sin 10}\cos(200^0 + 7.10) + \frac{1}{2} - \frac{1}{2} =$

$$= \frac{\sin 80}{\sin 10}\cos 270^0 = 0$$

63. Let $AP = x, PC = y$ & $PB = z$



$AB = \sqrt{x^2 + z^2}; AC = \sqrt{x^2 + y^2}$ & $BC = \sqrt{y^2 + z^2}$

$S = x + y + z + \sqrt{x^2 + y^2} + \sqrt{y^2 + z^2} + \sqrt{z^2 + x^2}$

Use $AM \geq GM$

$\Rightarrow x + y + z \geq 3(xyz)^{1/3}$ …….(1)

$\sqrt{x^2 + y^2} \geq \sqrt{2xy}$ ……..(2)

$\sqrt{y^2 + z^2} \geq \sqrt{2yz}$ ……..(3)

$\sqrt{z^2 + x^2} \geq \sqrt{2zx}$ …….(4)

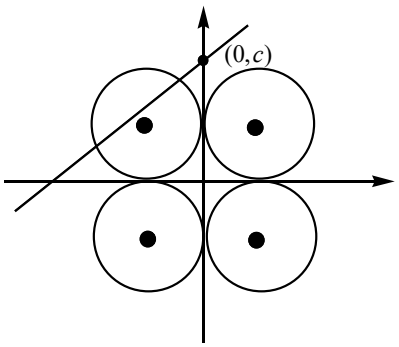
(1) + (2) + (3) + (4)

$\Rightarrow S \geq (xyz)^{1/3} + \sqrt{2}(\sqrt{xy} + \sqrt{yz} + \sqrt{zx})$

$\Rightarrow S \geq 3(1 + \sqrt{2})(xyz)^{1/3}$

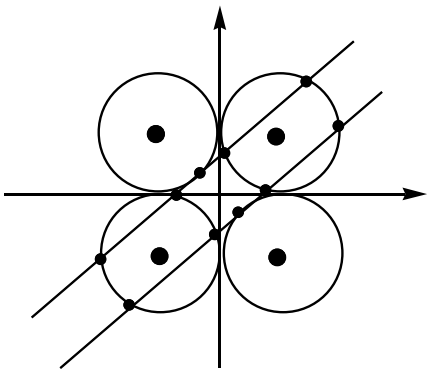
$\Rightarrow V = \frac{1}{6}xyz$

64. **i)**



$-2 \leq c \leq 2$

ii)



Clearly 'c' has two cases.

65. Clearly $a_1 = 2, a_2 = 3$ (00,01,10)

Now there are two mutually exclusive cases for the first box

If first box filled by zero

$\underline{0}_____$ then remaining (n -1) boxes can be filled by a_{n-1} ways.

If first box filled by 1 then 0 must be on the second box

$\therefore$ $\underline{0}\,\underline{1}\,_\,_\,_\,_\,_$ so remaining (n -2) boxes can be filled by a_{n-2} ways.

Therefore required recurrence relation is

$a_n = a_{n-1} + a_{n-2}$ …(1)

$$\Rightarrow \frac{a_n - a_{n-1}}{a_{n-2}} = 1$$

Also, $\left(\frac{a_5 a_6 - {a_5}^2}{a_4}\right) = a_5\left(\frac{a_6 - a_5}{a_4}\right) = a_5$

Using recurrence relation.

$a_5 = a_4 + a_3 = 2a_3 + a_2 = 2(a_2 + a_1) + a_2$

$= 3a_2 + 2a_1 = 3\times 3 + 2\times 2 = 13$

$\therefore$ Given expression $= 13 - 5\times 1 = 8$

66. Let point of intersection of curves is (x_1, y_1)

and let $2\cot\sqrt{\frac{\pi^2}{16} - \alpha^2} = k$

Now differentiate $\log(x^2 + y^2) = k\ \tan^{-1}\frac{y}{x}$

w.r.t. x we get $\frac{1}{x^2 + y^2}\left(2x + 2y\frac{dy}{dx}\right) = k\left(\frac{1}{1 + \frac{y^2}{x^2}}\right)\left\{\frac{x\frac{dy}{dx} - y}{x^2}\right\}$

$$\Rightarrow 2\left\{x + y\frac{dy}{dx}\right\} = k\left\{x\frac{dy}{dx} - y\right\} \Rightarrow \frac{dy}{dx} = \frac{2x + ky}{kx - 2y}$$

$$\Rightarrow \left.\frac{dy}{dx}\right]_{(x_1, y_1)} = \frac{2x_1 + ky_1}{kx_1 - 2y_1} = m_1$$

$\frac{dy}{dx}$ for $y = mx$ is $m = \frac{y_1}{x_1}$

$$\Rightarrow \tan\theta = \left|\frac{\frac{2x_1 + ky_1}{kx_1 - 2y_1} - \frac{y_1}{x_1}}{1 + \left(\frac{2x_1 + ky_1}{kx_1 - 2y_1}\right)\left(\frac{y_1}{x_1}\right)}\right| = \left|\frac{2x_1^2 + kx_1y_1 - kx_1y_1 + 2y_1^2}{kx_1^2 - 2x_1y_1 + 2x_1y_1 + ky_1^2}\right| = \left|\frac{2}{k}\right|$$

$$\Rightarrow \tan\theta = \tan\sqrt{\frac{\pi^2}{16} - \alpha^2} \Rightarrow \theta = \sqrt{\frac{\pi}{16} - \alpha^2}$$

$\Rightarrow$ max value of θ is $\frac{\pi}{4}$

$$\Rightarrow b = \frac{1}{4} = 0.25$$

67. Because discriminant of $x^2 + (2-a)x + 3$ is negative so $x^2 + (2-a)x + 3 = 0$ has both imaginary roots

$\Rightarrow$ both the equation have two common roots

Now, roots of equation $x^2 + (2-a)x + 3 = 0$ are

roots of $(ax^4 + bx^3 + x^2 + (3-a)x + 3) - (x^2 + (2-a)x + 3) = 0$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$ are

roots of $ax^4 + bx^3 + x = 0$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$

roots of $ax^3 + bx^2 + 1 = 0$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$ are

roots of $(ax^3 + bx^2 + 1) - (ax(x^2 + (2-a)x + 3)) = 0$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$ and

$(b - 2a + a^2)x^2 - 3ax + 1 = 0$ are same

$$\Rightarrow \frac{1}{b - 2a + a^2} = \frac{2-a}{-3a} = \frac{3}{1}$$

$$\Rightarrow a = -\frac{1}{4} \text{ and } b = -\frac{11}{48}$$

$$\Rightarrow |a - 12b| = 2.5$$

68. Let $(2\sec\theta, \tan\theta)$ be the point on hyperbola $\frac{x^2}{4} - = \frac{y^2}{1} = 1$, at which the tangent is being drawn

Equation of tangent: $\frac{x\sec\theta}{2} - y\tan\theta = 1$ .....(1)

Let (h,k) be the mid-point of chord PQ of the given ellipse

Equation of chord with given mid-point: $T = S_1$

$$\Rightarrow \frac{hx}{4} + \frac{ky}{1} = \frac{h^2}{4} + \frac{k^2}{1} \qquad \ldots.(2)$$

Comparing (1) and (2)

$$\Rightarrow \frac{\left(\frac{\sec\theta}{2}\right)}{\left(\frac{h}{4}\right)} = \frac{-\tan\theta}{k} = \frac{1}{\left(\frac{h^2}{4} + k^2\right)}$$

$$\Rightarrow \sec\theta = \frac{2h}{h^2 + 4k^2} \text{ and } \tan\theta = \frac{-4k}{h^2 + 4k^2}$$

$$\Rightarrow \frac{4h^2}{(h^2 + 4k^2)^2} - \frac{16k^2}{(h^2 + 4k^2)^2} = 1$$

$$\Rightarrow 4h^2 - 16k^2 = (h^2 + 4k^2)^2$$

Locus of mid-point of PQ: $(x^2 + 4y^2)^2 = 4(x^2 - 4y^2)$

Point of intersection with x-axis $\Rightarrow y = 0$

$$\Rightarrow x^4 = 4x^2 \Rightarrow x = 0, 2, -2$$

But (0,0) (2,0) and (-2,0) can not be mid point of any chord

69. $(x^2 - 10x + 9) \le 0 \Rightarrow (x-1)(x-9) \le 0$

$\Rightarrow x \in [1,9] \Rightarrow A = \{1,2,3,4,5,6,7,8,9\}$

$f(x) \le (x-3)^2 + 1$

$x = 1 : f(1) \le 5 \Rightarrow 1^2, 2^2$

$x = 2 : f(2) \le 2 \Rightarrow 1^2$

$x = 3 : f(3) \le 1 \Rightarrow 1^2$

$x = 4 : f(4) \le 2 \Rightarrow 1^2$

$x = 5 : f(5) \le 5 \Rightarrow 1^2, 2^2$

$x = 6 : f(6) \le 10 \Rightarrow 1^2, 2^2, 3^2$

$x = 7 : f(7) \le 17 \Rightarrow 1^2, 2^2, 3^2, 4^4$

$x = 8 : f(8) \le 26 \Rightarrow 1^2, 2^2, 3^2, 4^2, 5^2$

$x = 9 : f(9) \le 37 \Rightarrow 1^2, 2^2, 3^2, 4^2, 5^2, 6^2$

Total number of such function $= 2(6!) = 2(720) = 1440$